

Relations

Let A & B be two non-empty sets. A relation R from A to B is any subset of the Cartesian product $A \times B$ i.e.
 $R \subseteq A \times B$.

- If $(a, b) \in R$, we write aRb (read as "a is related to b").
- If $(a, b) \notin R$, a is not related to b.

Numericals:

Q.1) Let $A = \{1, 2, 3, 4\}$ & $B = \{1, 2\}$. Find the relation R from A to B such that aRb if a & b are both even numbers?

Solⁿ:

$$A \times B = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 2), (4, 1), (4, 2)\}$$

$$aRb = \{(2, 2), (4, 2)\}.$$

Q.2) Let $A = \{2, 4, 6, 8\}$ & $B = \{1, 2, 3\}$.

(i) Find R such that aRb if $a \geq b$?

(ii) Find R such that aRb if $a > b$?

(iii) Find R such that aRb if $a < b$?

Solⁿ:

$$A \times B = \{(2, 1), (2, 2), (2, 3), (4, 1), (4, 2), (4, 3), (6, 1), (6, 2), (6, 3), (8, 1), (8, 2), (8, 3)\}$$

(i) aRb (if $a \geq b$) = $\{(2, 1), (2, 2), (4, 1), (4, 2), (4, 3), (6, 1), (6, 2), (6, 3), (8, 1), (8, 2), (8, 3)\}$

(ii) aRb (if $a > b$) = $\{(2, 1), (4, 1), (4, 2), (4, 3), (6, 1), (6, 2), (6, 3), (8, 1), (8, 2), (8, 3)\}$

(iii) aRb (if $a < b$) = $\{(2, 3)\}$.

Q.3) If $A = \{1, 2, 3, 4\}$ & $B = \{2, 3, 4\}$, R is defined on set A to B such that aRb if a is divisible by b . Find R ?

Solⁿ:

$$A \times B = \{(1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (3, 4), (4, 2), (4, 3), (4, 4)\}$$

$$aRb = \{(2, 2), (3, 3), (4, 2), (4, 4)\}$$

Q.4) Let $A = \{1, 2, 3\}$ - ~~Find~~ and $B = \{1, 2, 5\}$. Find R such that aRb if $a \times b$ is a prime number?

Solⁿ:

$$A \times B = \{(1, 1), (1, 2), (1, 5), (2, 1), (2, 2), (2, 5), (3, 1), (3, 2), (3, 5)\}$$

$$aRb = \{(1, 2), (1, 5), (2, 1), (3, 1)\}$$

Q.5) $A = \{1, 3, 5\}$, $B = \{1, 2, 3, 4, 5, 6\}$ and $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Find A^c , B^c , $(A \cap B)^c$, $(A^c \cap B^c)$, $(A \cup B^c)$, $[A - (A^c - B^c)] \& [B \oplus (B^c - A)]$?

Solⁿ:

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{1, 3, 5\}$$

$$B = \{1, 2, 3, 4, 5, 6\}$$

$$A^c = \{2, 4, 6, 7, 8, 9, 10\}$$

$$B^c = \{7, 8, 9, 10\}$$

$$A \cap B = \{1, 3, 5\} \Rightarrow (A \cap B)^c = \{2, 4, 6, 7, 8, 9, 10\}$$

$$A^c \cap B^c = \{7, 8, 9, 10\}$$

$$A \cup B^c = \{1, 3, 5, 7, 8, 9, 10\}$$

$$A - (A^c - B^c) = \{1, 3, 5\} \quad [A^c - B^c = \{2, 4, 6\}]$$

$$B \oplus (B^c - A) = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

Types of Relation

1. Universal Relation:

Relation R on set A is called universal relation if $R = A \times A$.

$$A = \{1, 2, 3\}$$

$$R = A \times A = \{(1, 1), (2, 2), (3, 3), (1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 2)\}.$$

2. Identity Relation:

Relation R on set A is called identity relation if every element of set is related to itself i.e. aRa .

$$A = \{1, 2\}$$

$$R = \{(1, 1), (2, 2)\}.$$

3. Empty Relation:

Relation R on set A is called empty relation if $R = \emptyset$.

$$A = \{1, 2, 3\}$$

$$R (\text{if } a+b > 8) = \{\emptyset\}.$$

4. Reflexive Relation:

Relation R on set A is called reflexive relation if aRa is $(a,a) \in R$ for every element $a \in A$ i.e. $\forall a ((a,a) \in R)$.

$$A = \{1, 2, 3, 4\}$$

$$\text{Reflexive, } R_1 = \{(1,1), (1,2), (2,2), (2,3), (3,3), (4,4)\}$$

$$\text{Non-Reflexive, } R_2 = \{(1,1), (1,2), (2,1), (2,2), (3,1), (4,4)\}.$$

5. Irreflexive Relation:

Relation R on set A is called irreflexive relation if aRa is $(a,a) \notin R$ for every element $a \in A$.

$$A = \{1, 2, 3, 4\}$$

$$R = \{(1,2), (2,3)\}$$

6. Symmetric Relation:

Relation R on set A is said to be symmetric if $(a,b) \in R \Rightarrow (b,a) \in R$.

$$A = \{1, 2, 3\}$$

$$R = \{(1,1), (1,2), (2,1), (2,3), (3,2)\}.$$

7. Asymmetric Relation:

Relation R on set A is said to be asymmetric if $(a, b) \in R \Rightarrow (b, a) \notin R$.

$$A = \{1, 2, 3\}$$

$$R = \{(1, 2), (2, 3)\}.$$

8. Anti-symmetric Relation:

Relation R on set A is antisymmetric if $(a, b) \in R$ & $(a, b) \in R \Rightarrow (b, a) \in R, a = b$.

$$A = \{1, 2, 3\}$$

$$R_1 = \{(1, 1), (2, 2)\}$$

(Anti-symmetric)

$$A = \{4, 5, 6\}$$

$$R_2 = \{(4, 4), (4, 5), (5, 4), (5, 6), (4, 6)\}.$$

(Not anti-symmetric)

9. Transitive Relation:

Relations R on set A is transitive relation if $(a, b) \in R$ and $(b, c) \in R$ then $(a, c) \in R$.

$$A = \{1, 2, 3\}$$

$$\text{Transitive, } R_1 = \{(1, 2), (2, 3), (1, 3)\}$$

$$\text{Non-transitive, } R_2 = \{(1, 3), (3, 2), (2, 2), (1, 1)\}.$$

Compliment of Relation

Let R be a relation from A to B then the compliment of relation will be R^c or R' consists of all ordered pairs in the Cartesian product $A \times B$ that are not present in R .

$$\text{i.e. } R^c = \{(a, b) \mid (a, b) \in A \times B \text{ and } (a, b) \notin R\}$$

$$\text{Ex. } A \times B = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$$

$$R = \{(a, 1), (a, 3), (b, 3)\}$$

$$R^c = \{(a, 2), (b, 1), (b, 2)\}.$$

Inverse of Relation

Let R be a relation from A to B then the inverse of relation will be a relation from B to A . It is denoted by R^{-1} .

$$A \times B = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$$

$$R = \{(a, 1), (a, 3), (b, 3)\}$$

$$R^{-1} = \{(1, a), (3, a), (3, b)\}.$$

Composite Relation

Let A, B & C be three sets and R be there relation from A to B and S be relation from B to C .

$$R \subseteq A \times B$$

$$\& S \subseteq B \times C.$$

Then, the composition of R & S can be fixed as $(R.S \text{ or } R \circ S)$,

$$R.S = \{(a, c) : \text{there exists } b \in B \text{ for which } (a, b) \in R \text{ and } (b, c) \in S\}.$$

Example:

$$A = \{1, 2, 3, 4\}, B = \{a, b, c, d\} \& C = \{x, y, z\}$$

$$R = \{(1, a), (2, d), (3, a), (3, b), (3, d)\}$$

$$S = \{(b, x), (b, z), (c, y), (d, z)\}$$

$$R.S = \{(2, z), (3, x), (3, z)\}.$$

$$S.R = \emptyset$$

$$R.R = \emptyset$$

$$S.S = \emptyset.$$

Q. Let, $A = \{1, 3, 5, 7\}$ and $B = \{1, 2, 4, 8\}$. Find:

- (i) The relation R_1 defined by $a \leq b$ (where $a \in A$ and $b \in B$).
- (ii) The relation R_2 defined by $a \times b = \text{odd numbers}$.
- (iii) Find R_1^c , R_1^{-1} , R_2^c & R_2^{-1} .

Solⁿ:

$$A \times B = \{(1, 1), (1, 2), (1, 4), (1, 8), (3, 1), (3, 2), (3, 4), (3, 8), (5, 1), (5, 2), (5, 4), (5, 8), (7, 1), (7, 2), (7, 4), (7, 8)\}$$

$$R_1 = \{(1, 1), (1, 2), (1, 4), (1, 8), (3, 4), (3, 8), (5, 8), (7, 8)\}$$

$$R_1^c = \{(3, 1), (3, 2), (5, 1), (5, 2), (5, 4), (7, 1), (7, 2), (7, 4)\}$$

$$R_1^{-1} = \{(1, 1), (2, 1), (4, 1), (8, 1), (4, 3), (8, 3), (8, 5), (8, 7)\}$$

Also,

$$R_2 = \{(1, 1), (3, 1), (5, 1), (7, 1)\}$$

$$R_2^c = \{(1, 2), (1, 4), (1, 8), (3, 2), (3, 4), (3, 8), (5, 2), (5, 4), (5, 8), (7, 2), (7, 4), (7, 8)\}$$

$$R_2^{-1} = \{(1, 1), (1, 3), (1, 5), (1, 7)\}$$

Q. Let, $R = \{(a, b), (c, d), (b, b)\}$ and
 $S = \{(d, b), (b, e), (c, a), (a, c)\}$.

Find $R.S$, $S.R$, $(R.S).R$, $(S.R).R$, $R.R$, $R.R.R$,
 $S.S$, $R.(S.R)$?

Solⁿ:

$$R.S = \{(a, e), (c, b), (b, e)\}$$

$$S.R = \{(d, b), (c, b), (a, d)\}$$

$$(R.S).R = \{(c, b)\}$$

$$(S.R).R = \{(d, b), (c, b)\}$$

$$R.R = \{(a, b), (b, b)\}$$

$$R.R.R = \{(a, b), (b, b)\}$$

$$S.S = \{(d, e), (c, c), (a, a)\}$$

$$R.(S.R) = \{(c, b)\}.$$

Q. Let $A = \{1, 2, 3\}$ and $B = \{2, 3, 4\}$.
Find $(A \times B) - (B \times A)$?

Solⁿ:

$$A = \{1, 2, 3\}$$

$$\& B = \{2, 3, 4\}$$

Now,

$$A \times B = \{(1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (3, 4)\}$$

Also,

$$B \times A = \{(2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3), (4, 1), (4, 2), (4, 3)\}$$

$$\therefore (A \times B) - (B \times A) = \{(1, 2), (1, 3), (1, 4), (2, 4), (3, 4)\}.$$

Transitive Closure Using Warshall's Algorithm

Q.1) Using Warshall's Algorithm, find transitive closure of the relation R on set A ?

$$R = \{(1,1), (1,3), (2,2), (3,1), (3,2)\}$$

$$\& A = \{1, 2, 3\}$$

Solⁿ:

$$M_R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

	I st	II nd	III rd
Column	$\{1, 3\}$	$\{2, 3\}$	$\{1\}$
Row	$\{1, 3\}$	$\{2\}$	$\{1, 2\}$
Column X Row	$\{(1,1), (1,3), (3,1), (3,3)\}$	$\{(2,2), (3,2)\}$	$\{(1,1), (1,2)\}$

$$R_1 = \{(1,1), (1,2), (1,3), (2,2), (3,1), (3,2), (3,3)\}$$

Transitive closure,
 $R_2 = R_1 \cup R$

$$\Rightarrow R_2 = \{(1,1), (1,2), (1,3), (2,2), (3,1), (3,2), (3,3)\}.$$

Q.2) Find transitive closure of the relation R on set A ?

$$R = \{(a, d), (b, a), (b, c), (c, a), (c, d), (d, c)\}$$

$$A = \{a, b, c, d\}$$

Solⁿ:

	a	b	c	d
a	0 ₀	0	0 ₀	1
b	1	0	1	0 ₀
c	1	0	0 ₀	1
d	0 ₀	0	1	0 ₀

	I st	II nd	III rd	IV th
Column	$\{b, c\}$	ϕ	$\{b, d\}$	$\{a, b, c, d\}$
Row	$\{d\}$	$\{a, c, d\}$	$\{a, d\}$	$\{a, c, d\}$
Column X Row	$\{(b, d), (c, d)\}$	ϕ	$\{(b, a), (b, d), (d, a), (d, d)\}$	$\{(a, a), (a, c), (a, d), (b, a), (b, c), (b, d), (c, a), (c, c), (c, d), (d, a), (d, c), (d, d)\}$

$$R_1 = \{(a, a), (a, c), (a, d), (b, a), (b, c), (b, d), (c, a), (c, c), (c, d), (d, a), (d, c), (d, d)\}$$

Q.3) Find transitive closure of the relation R on set A ?

$$R = \{(1, 4), (2, 1), (2, 3), (3, 1), (4, 3), (3, 4)\}$$

$$A = \{1, 2, 3, 4\}$$

Solⁿ:

	1	2	3	4
1	0 ₀	0	0 ₀	1
2	1	0	1	0 ₀
3	1	0	0 ₀	1
4	0 ₀	0	1	0 ₀

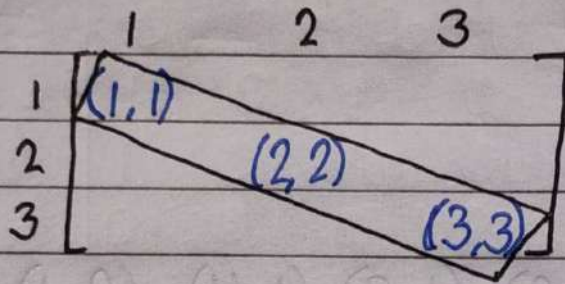
	I st	II nd	III rd	IV th
Column	$\{2, 3\}$	ϕ	$\{2, 4\}$	$\{1, 2, 3, 4\}$
Row	$\{4\}$	$\{1, 3, 4\}$	$\{1, 4\}$	$\{1, 3, 4\}$
Column X Row	$\{(2, 4), (3, 4)\}$	ϕ	$\{(2, 1), (2, 4), (4, 1), (4, 4)\}$	$\{(1, 1), (1, 3), (1, 4), (2, 1), (2, 3), (2, 4), (3, 1), (3, 3), (3, 4), (4, 1), (4, 3), (4, 4)\}$

$$R_1 = \{(1, 1), (1, 3), (1, 4), (2, 1), (2, 3), (2, 4), (3, 1), (3, 3), (3, 4), (4, 1), (4, 3), (4, 4)\}.$$

Diagonal Relation

A relation R on set A is said to be diagonal relation if R is the set of all ordered pairs $(x, x) \forall x \in A$ and it is denoted by Δ_+ .

Example: $R = \{(x, x) \forall x \in A\}$



Ordered of Relation

An order of relation is a binary relation that describes a logical sequence among elements in a set.

If a relation R provides us with a method to rank elements of a set against one another, it is called an ordered relation.

There are three types of ordered relations, they are:

1. Partial Order Relation
2. Total or Linear Order Relation
3. Quasi Order Relation

1. Partial Order Relation:

Any order relation on set A is partial order if it is reflexive, antisymmetric and transitive.

Q. Show that the relation $(a, b) \in R$ iff $a \geq b$ define on set of integer (+ve) is a partial order relation?

$$A = \{1, 2, 3, 4\}$$

$$A \times A = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (2, 4), (3, 1), (3, 2), (3, 3), (3, 4), (4, 1), (4, 2), (4, 3), (4, 4)\}$$

~~Reflexive~~ $R = \{(1, 1), (2, 1), (2, 2), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3), (4, 4), (3, 3)\}$

Reflexive, $R_1 = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$
 $[\forall a \in A, (a, a) \in R]$

Anti-symmetric, $R_2 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$
 $\left[\begin{array}{l} \forall a, b \in A \\ \text{if } (a, b) \in R \ \& \ (b, a) \in R \\ \text{then } a = b \end{array} \right]$

Transitive, $R_3 = \{(3, 1), (4, 2), (4, 1)\}$
 $\left[\begin{array}{l} \forall a, b, c \in A, \\ (a, b) \in R \ \& \ (b, c) \in R, \\ \text{then } (a, c) \in R \end{array} \right]$

Q. Show that the relation $(a, b) \in R$ iff $a > b$ define on set of positive integer is a partial order relation?

$$A = \{1, 2, 3, 4\}$$

$$R = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$$

$$\text{Reflexive, } R_1 = \emptyset$$

Since, $\forall a \in A, (a, a) \notin R$ so relation is not reflexive.

Hence, R is not a partial order relation.

2. Total or Linear Order Relation:

A partial order is said to be total order relation if order provide a method of comparison between two elements in a set.

$$\forall a, b \in A \text{ either } aRb \text{ or } bRa.$$

Example: Check that relation $(a, b) \in R$ if $a \leq b$ define on set of positive integer is total order relation.

$$A = \{5, 6, 7, 8\}$$

$$A \times A = \{(5, 5), (5, 6), (5, 7), (5, 8), (6, 5), (6, 6), (6, 7), (6, 8), (7, 5), (7, 6), (7, 7), (7, 8), (8, 5), (8, 6), (8, 7), (8, 8)\}$$

$$R = \{(5, 5), (5, 6), (5, 7), (5, 8), (6, 6), (6, 7), (6, 8), (7, 7), (7, 8), (8, 8)\}$$

$$\text{Reflexive, } R_1 = \{(5, 5), (6, 6), (7, 7), (8, 8)\}$$

$$\text{Anti-symmetric, } R_2 = \{(5, 6), (5, 7), (5, 8), (6, 7), (6, 8), (7, 8)\}$$

$$\text{Transitive, } R_3 = \{(5, 7), (5, 8), (6, 8)\}.$$

Example: Check that relation $(a, b) \in R$ if a/b define on set of positive integer is total order relation.

$$A = \{5, 6, 7, 8\}$$

$$R = \{(5, 5), (6, 6), (7, 7), (8, 8)\}$$

Reflexive, $R_1 = \{(5, 5), (6, 6), (7, 7), (8, 8)\}$

Anti-symmetric, $R_2 = \emptyset$

Transitive, $R_3 = \emptyset$.

Hence, the relation is not a total order relation.

3. Quasi Order Relation:

A relation R on set A is quasi order relation iff it is irreflexive and transitive.

Example: Show that ' $<$ ' less than relation on set of positive integer is Quasi-order relation.

$$A = \{1, 2, 3, 4\}$$

$$A \times A = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (2, 1), (2, 2), (3, 1), (3, 2), (3, 3), (3, 4), (4, 1), (4, 2), (4, 3), (4, 4)\}$$

$$R = \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$$

Irreflexive, if aRa is $(a, a) \notin R$, which is true.

Transitive, if $(a, b) \in R$ & $(b, c) \in R$ then $(a, c) \in R$,
 $(1, 2) \in R$ & $(2, 3) \in R \Rightarrow (1, 3) \in R$
 $(1, 2) \in R$ & $(2, 4) \in R \Rightarrow (1, 4) \in R$
 $(1, 3) \in R$ & $(3, 4) \in R \Rightarrow (1, 4) \in R$
 $(2, 3) \in R$ & $(3, 4) \in R \Rightarrow (2, 4) \in R$

So, R is transitive.

Hence, the relation R is Quasi-order relation.

Equivalence Class

Equivalence class of x is denoted by,
 $|X| = \{y \mid y \in A \text{ \& } (x,y) \in R\}.$

When a relation is equivalence then we can find the equivalence class.

Example:

$$A = \{1, 2, 3, 4, 5\}$$

$$R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (2,1), (4,5), (5,4)\}$$

Equivalence Class,

$$[1] = \{1, 2\} \quad P_1 \text{ (Partition)}$$

$$[2] = \{2, 1\} \text{ or } \{1, 2\} \quad P_1$$

$$[3] = \{3\} \quad P_2$$

$$[4] = \{4, 5\} \quad P_3$$

$$[5] = \{5, 4\} \text{ or } \{4, 5\} \quad P_3$$

Note: $P_1 \cup P_2 \cup P_3 = A$

$$\& \quad P_1 \cap P_2 \cap P_3 = \emptyset.$$

Partial Order Set (POSET)

A relation R on a set A is said to be partial order relation, if it is reflexive, anti-symmetric and transitive.

1. Reflexive, $aRa \forall a \in A$.
2. Anti-symmetric, $\textcircled{i} (a, b) \in R$
OR $\textcircled{ii} (a, b) \in R \ \& \ (b, a) \in R \Rightarrow a = b$
3. Transitive, $aRb \ \& \ bRc \Rightarrow aRc$.

Then the given set is partial order set i.e. POSET.

Strict Partial Order Set (POSET):

A relation R on a set A is said to be strict POSET, if the relation is irreflexive, anti-symmetric and transitive.

Irreflexive, $(a, a) \notin R$ or aRa .

Then the given set is strict POSET.

Q. Check the following sets are partial order sets or not.

1. $A = \{1, 2, 3\}$
 $R = \emptyset$

Solⁿ:

Since, R is not partial order relation, so set is not POSET.

2. $A = \{1, 2, 3\}$
 $R = \{(1, 1), (2, 2), (3, 3)\}$

Here, R is reflexive, anti-symmetric and transitive.

So, set is POSET.

3. $A = \{1, 2, 3\}$
 $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$

Here, R is not anti-symmetric.

So, set is not POSET.

4. $A = \{1, 2, 3\}$
 $R = \{(1, 1), (2, 2), (3, 3), (1, 3), (2, 3)\}$

Here, R is partial order relation.

So, set is POSET.